RAMON MAGSAYSAY MEMORIAL COLLEGES
Information Technology Education Program
1st SEMESTER: AY: 2025 - 2026

NAME: _____ SCHEDULE: _____ SCORE: _____
SUBJECT: **WEB SYSTEMS AND TECHNOLOGIES** INSTRUCTOR: _____ DATE: _____

LABORATORY EXERCISE 1
SETTING UP THE DEVELOPMENT ENVIRONMENT & YOUR FIRST HTML PAGE

Learning Objectives

By the end of this laboratory exercise, students should be able to:

- Successfully install and configure XAMPP to run a local Apache web server.
- Install and set up Visual Studio Code (VS Code) as a code editor.
- Install the "Live Server" extension to streamline the development workflow.
- Create a structured project folder inside the server's document root (htdocs).
- Write a basic HTML5 document using the standard boilerplate structure.
- View their first webpage using both the localhost URL and the Live Server feature..

Prerequisite student experiences and knowledge

Before starting this exercise, students should have:

- None! This laboratory is designed for absolute beginners.
- Basic familiarity with using a computer (creating folders, downloading files, and navigating directories).
- Ability to use a web browser (Chrome, Firefox, or Edge).

Background

When you visit a website like facebook.com or youtube.com, your browser is actually requesting files from a remote computer called a web server. To practice web development, we don't need to rent a remote server. Instead, we install a local web server on our own PC.

XAMPP is a free software package that includes Apache (the web server). When you place your HTML files inside XAMPP's htdocs folder, Apache serves them to your browser via http://localhost. This perfectly simulates a live environment without needing the internet.

Visual Studio Code is a lightweight but powerful code editor. It provides syntax highlighting (color-coding your code) and extensions like "Live Server", which automatically refreshes your browser whenever you save your HTML file—saving you a huge amount of time.

In this exercise, you will set up both tools and write your very first HTML page from scratch.

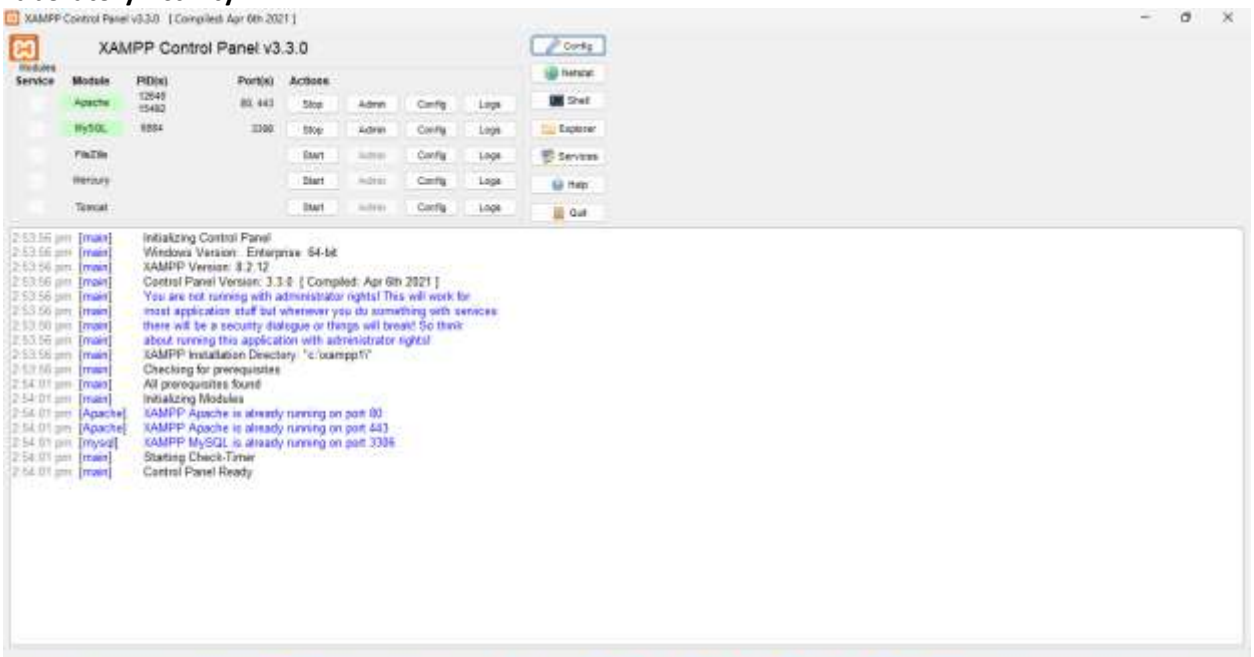
Materials/Resources

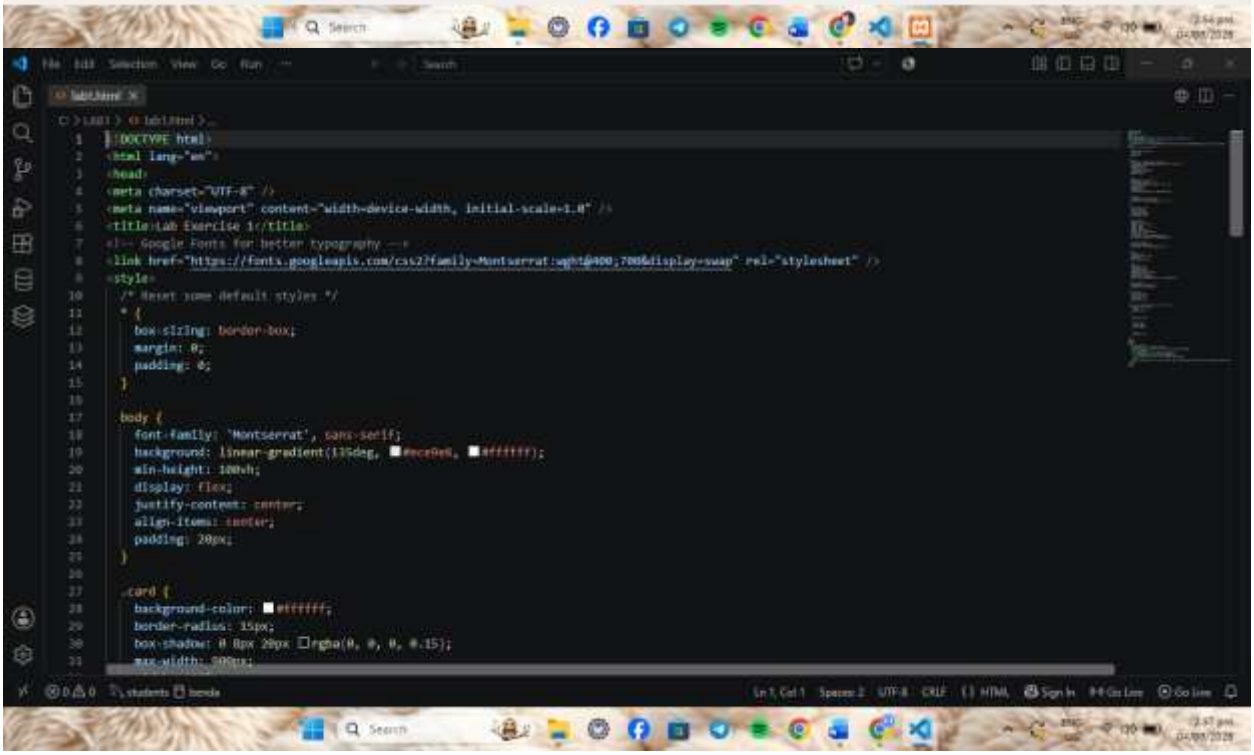
- Personal Computer with Internet Access (Windows, macOS, or Linux).
- XAMPP Installer (downloaded from apachefriends.org).
- Visual Studio Code (downloaded from code.visualstudio.com).
- Web Browser (Google Chrome is recommended).
- Github (https://github.com/)

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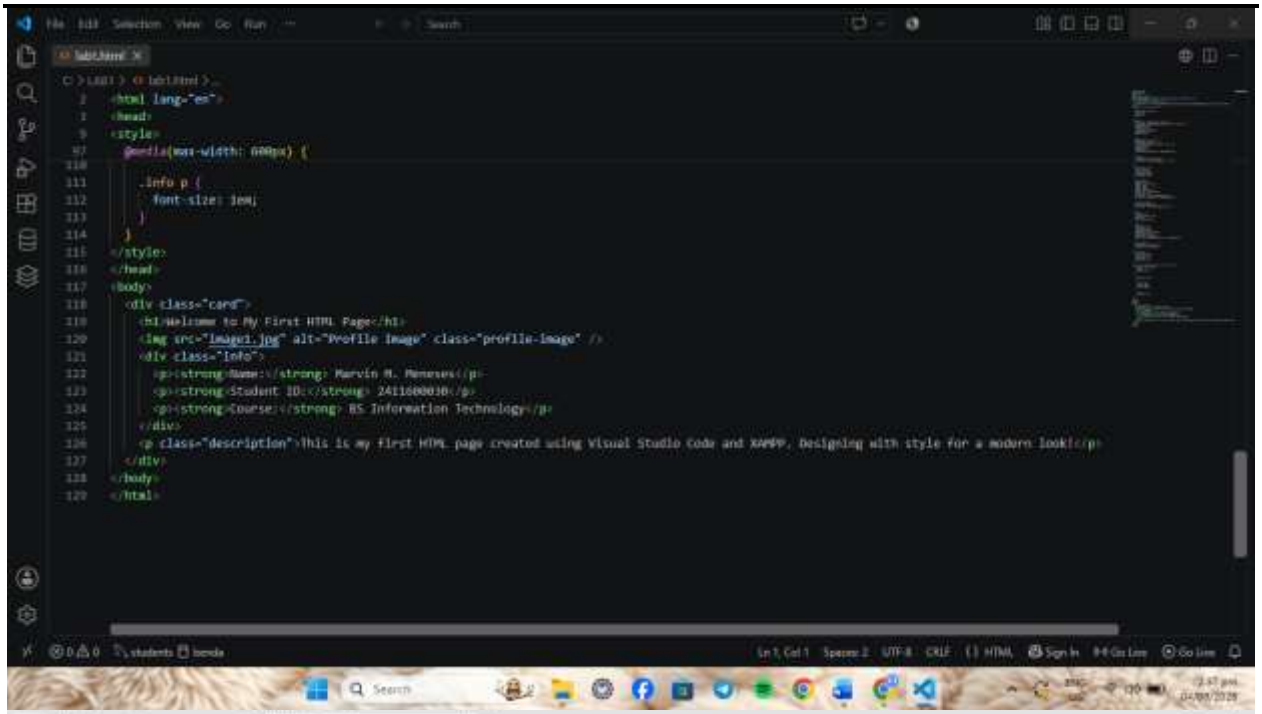
Laboratory Activity



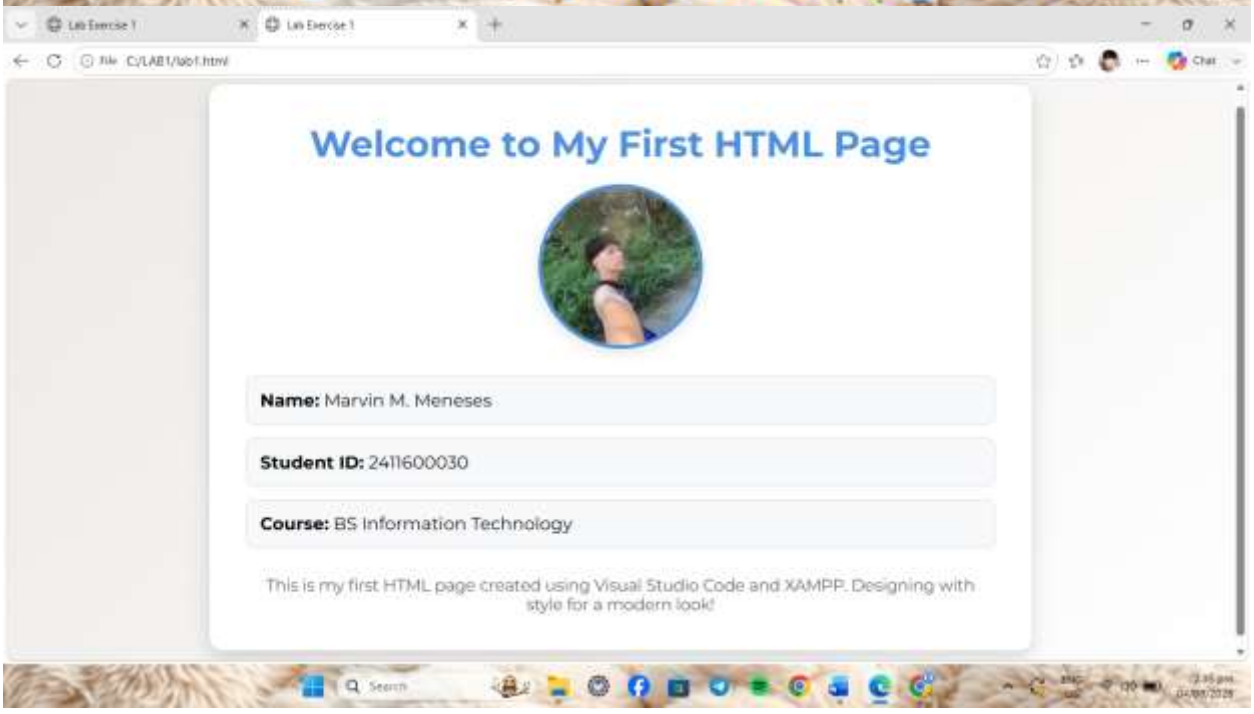


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```
<!DOCTYPE html>
<html lang="en">
<head>
<style>
  @media(max-width: 600px) {
    .info p {
      font-size: 1em;
    }
  }
</style>
</head>
<body>
  <div class="card">
    <h1>Welcome to My First HTML Page</h1>
    <img src="image1.jpg" alt="Profile Image" class="profile-image" />
    <div class="info">
      <p><strong>Name:</strong> Marvin M. Meneses</p>
      <p><strong>Student ID:</strong> 2411600030</p>
      <p><strong>Course:</strong> BS Information Technology</p>
    </div>
    <p class="description">This is my first HTML page created using Visual Studio Code and XAMPP. Designing with style for a modern look!</p>
  </div>
</body>
</html>
```



Welcome to My First HTML Page

Name: Marvin M. Meneses

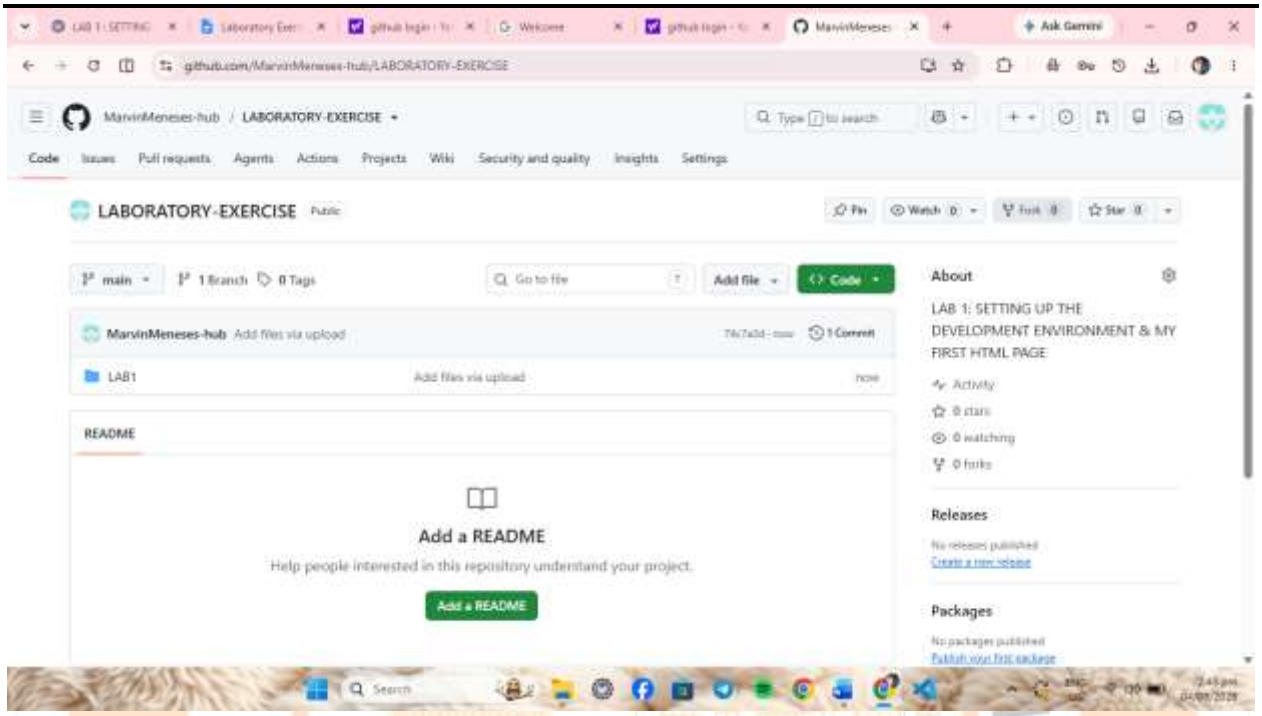
Student ID: 2411600030

Course: BS Information Technology

This is my first HTML page created using Visual Studio Code and XAMPP. Designing with style for a modern look!

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Submission Guidelines

- ✓ Name the file as: Lab1_[YourStudentID]_[YourLastName].pdf (e.g., Lab1_20251234_Cruz.pdf).
- ✓ Submit the File to your instructor via the Learning Management System (LMS) provided before the start of the next laboratory session.

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QUESTIONS:

1. What is the main difference between opening an HTML file directly via file:/// and opening it via http://localhost? Why is localhost preferred in web development?
2. What is the purpose of the htdocs folder in XAMPP, and what happens if you place your files outside of it?
3. If Apache refuses to start because Port 80 is in use, name two programs that might be causing the conflict and explain the step you took to resolve it.
4. What is the role of the <!DOCTYPE html> declaration at the very top of your HTML file?
5. How does the "Live Server" extension in VS Code improve a developer's workflow compared to manually refreshing the browser?

ANSWERS:

1. The main difference between opening an HTML file through file:/// and http://localhost is how the browser accesses the file. When I use file:///, the browser simply opens the HTML file directly from my computer without using a web server. This works for basic HTML and CSS, but some features, such as PHP, databases, and certain JavaScript functions, may not work because they require a server. On the other hand, http://localhost loads the website through a local web server like Apache in XAMPP. This is the preferred method in web development because it closely simulates how a website works when it is hosted online, making it easier to test and identify issues before deployment.

2. The htdocs folder in XAMPP serves as the default location where Apache stores and accesses website files. Whenever I want to view my project using http://localhost, I need to save the project inside the htdocs folder so that Apache can find and serve it. If I place my files somewhere outside of htdocs, the browser will not be able to access them through localhost because Apache does not automatically search other folders. Unless I manually change Apache's configuration to point to another directory, the project will not load correctly.

3. If Apache refuses to start because Port 80 is already being used, it usually means another program is using that port. Two common examples are Internet Information Services (IIS), which is built into Windows, and Skype, which may use Port 80 in some configurations. To solve the problem, I can stop or disable the program that is using the port through the Windows Services or Task Manager. Another option is to change Apache's listening port from 80 to 8080 in its configuration settings. After restarting Apache, I can access my local website by typing http://localhost:8080 in the browser.

4. The <!DOCTYPE html> declaration is placed at the very top of an HTML document to tell the browser that the page is written using the HTML5 standard. Although it is not an HTML tag itself, it is very important because it ensures the browser displays the webpage in standards mode instead of quirks mode. This helps the page behave consistently across different web browsers

and reduces compatibility issues. Including this declaration is considered a best practice in modern web development.

5. The Live Server extension in Visual Studio Code makes web development much more convenient because it automatically refreshes the webpage whenever I save changes to my code. Instead of manually refreshing the browser every time I edit an HTML, CSS, or JavaScript file, the browser updates instantly, allowing me to immediately see the results of my changes. This saves time, speeds up the testing process, and makes it easier to identify and fix mistakes while working on a project. Overall, Live Server creates a smoother and more efficient workflow for developers.

Output / Results

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Conclusion

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